

1.

```
GNU nano 8.7.1 practical.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);

    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());

        execlp(cmd, cmd, NULL);

        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());

        wait(NULL);

        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }

    return 0;
}
```

```
varshithreddy@Sai:~$ nano practical.c
varshithreddy@Sai:~$ gcc practical.c -o practical
varshithreddy@Sai:~$ ./practical.c
-bash: ./practical.c: Permission denied
varshithreddy@Sai:~$ practical
practical: command not found
varshithreddy@Sai:~$ ./practical
Enter Linux command: s
Parent PID : 978
Child PID : 979
Execution Failed: No such file or directory
Child process completed.
```

```
varshithreddy@Sai:~$ ./practical
Enter Linux command: uname
Parent PID : 1042
Child PID : 1043
Linux
Child process completed.
varshithreddy@Sai:~$ |
```

```

varshithreddy@Sai:~$ nano practical.c
varshithreddy@Sai:~$ uname
Linux
varshithreddy@Sai:~$ ./practical
Enter Linux command: lscpu
Parent PID : 1025
Child PID : 1026
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:         46 bits physical, 48 bits virtual
Byte Order:            Little Endian
CPU(s):                16
On-line CPU(s) list:   0-15
Vendor ID:             GenuineIntel
Model name:            Intel(R) Core(TM) Ultra 9 285H
CPU family:            6
Model:                 197
Thread(s) per core:    1
Core(s) per socket:    16
Socket(s):             1
Stepping:              2
BogoMIPS:              7372.79
Flags:                 fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
cp_lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
bbp stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflush
opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
clear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:        VT-x
Hypervisor vendor:     Microsoft
Virtualization type:   full
Caches (sum of all):
L1d:                   768 KiB (16 instances)
L1i:                   1 MiB (16 instances)
L2:                    48 MiB (16 instances)
L3:                    24 MiB (1 instance)
NUMA:
NUMA node(s):          1
NUMA node0 CPU(s):     0-15
Vulnerabilities:
Gather data sampling:   Not affected
Ghostwrite:            Not affected
Indirect target selection: Not affected
Itlb multihit:         Not affected
L1tf:                  Not affected
Mds:                   Not affected
Meltdown:              Not affected
Mmio stale data:       Not affected
Old microcode:         Not affected
Reg file data sampling: Not affected
Retbleed:              Mitigation; Enhanced IBRS
Spec rstack overflow:  Not affected
Spec store bypass:     Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:            Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds:                 Not affected
Tsx:                   Not affected
Tsx async abort:       Not affected
Vmscape:               Not affected
Child process completed.

```

```

varshithreddy@Sai:~$ ./practical
Enter Linux command: ps
Parent PID : 1105
Child PID : 1108
  PID TTY          TIME CMD
  844 pts/0        00:00:00 bash
  1105 pts/0        00:00:00 practical
  1108 pts/0        00:00:00 ps
Child process completed.
varshithreddy@Sai:~$ |

```

```
varshithreddy@Sai:~$ ./practical
Enter Linux command: top
Parent PID : 1120
Child PID : 1121
top - 09:26:32 up 31 min, 1 user, load average: 0.10, 0.05, 0.01
Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.6 sy, 0.0 ni, 99.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 15673.6 total, 14932.7 free, 648.1 used, 289.9 buff/cache
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used, 15025.6 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24180	15380	11552	S	0.0	0.1	0:00.72	systemd
2	root	20	0	3180	2200	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
8	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
48	root	19	-1	50416	16716	15444	S	0.0	0.1	0:00.22	systemd-journal
80	systemd+	20	0	22540	14624	12000	S	0.0	0.1	0:00.06	systemd-resolve
89	root	20	0	35524	12416	9152	S	0.0	0.1	0:00.91	systemd-udev
109	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.02	chronyd-starter
110	root	20	0	44772	3104	2852	S	0.0	0.0	0:00.01	cron
111	message+	20	0	8792	5396	4708	S	0.0	0.0	0:00.06	dbus-daemon
142	root	20	0	44092	29792	14648	S	0.0	0.2	0:00.21	networkd-dispat
174	root	20	0	18432	9468	8224	S	0.0	0.1	0:00.11	systemd-logind
193	syslog	20	0	220548	5940	4660	S	0.0	0.0	0:00.07	rsyslogd
222	_chrony	20	0	24540	12012	7156	S	0.0	0.1	0:00.16	chronyd
239	_chrony	20	0	12092	2468	1824	S	0.0	0.0	0:00.01	chronyd
295	root	20	0	123456	32540	17924	S	0.0	0.2	0:00.14	unattended-upgr
297	root	20	0	5492	2740	2556	S	0.0	0.0	0:00.00	agetty
343	root	20	0	8644	5540	4700	S	0.0	0.0	0:00.01	login
388	varshit+	20	0	22340	13604	10984	S	0.0	0.1	0:00.14	systemd
390	varshit+	20	0	22908	4088	2088	S	0.0	0.0	0:00.00	(sd-pam)
420	varshit+	20	0	6136	5396	3848	S	0.0	0.0	0:00.04	bash
836	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
838	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.15	Relay(844)
844	varshit+	20	0	6268	5532	3860	S	0.0	0.0	0:00.15	bash
1062	root	20	0	1866288	15468	12560	S	0.0	0.1	0:00.09	wsl-pro-service
1120	varshit+	20	0	2768	1880	1784	S	0.0	0.0	0:00.00	practical
1121	varshit+	20	0	8276	5724	3572	R	0.0	0.0	0:00.01	top

2.

```
GNU nano 8.7.1                                practicalla.
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

```
varshithreddy@Sai:~$ nano practicalla
varshithreddy@Sai:~$ gcc practicalla.c -o practicalla
varshithreddy@Sai:~$ ./practicalla
Cannot open source file
varshithreddy@Sai:~$ nano input.txt
varshithreddy@Sai:~$ ./practicalla
File copied successfully.
varshithreddy@Sai:~$ |
```